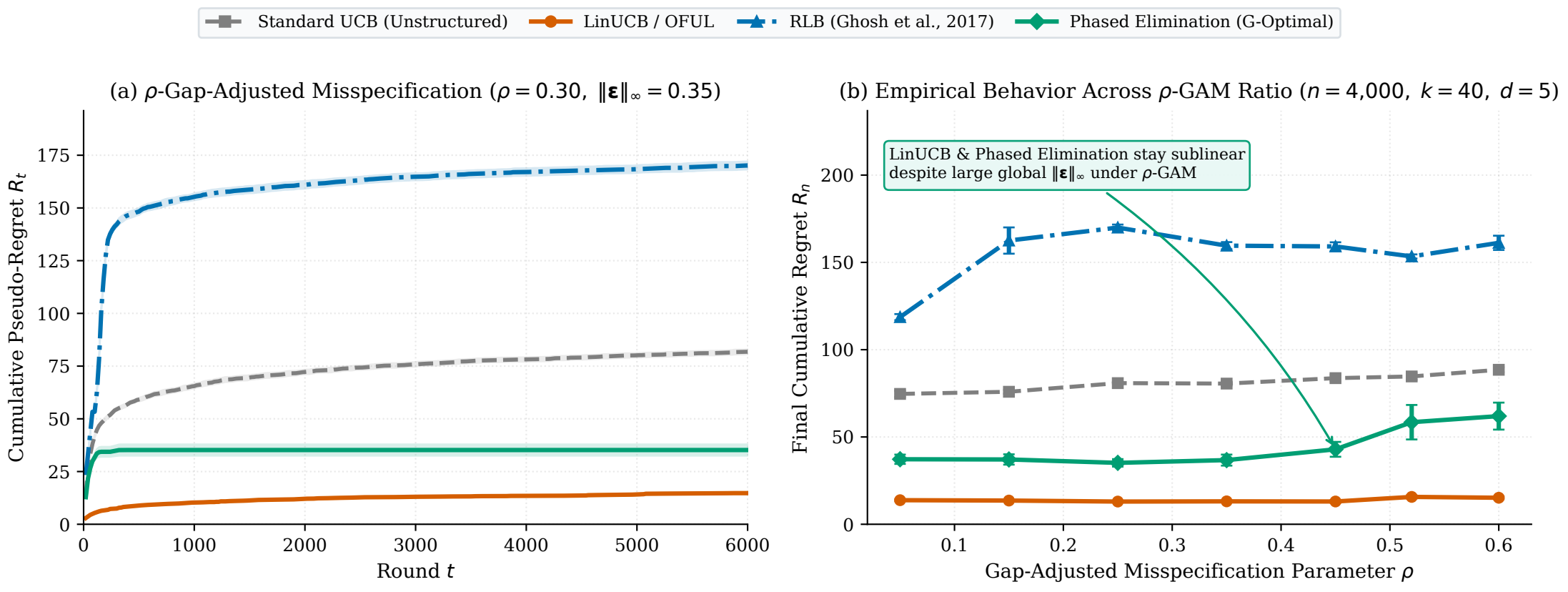
Standard UCB (Unstructured)
LinUCB / OFUL
RLB (Ghosh et al., 2017)
Phased Elimination (G-Optimal)
(a) ρ-Gap-Adjusted Misspecification (ρ = 0.30, ‖ε‖∞ = 0.35)
Cumulative Pseudo-Regret R_t
Round t
0
25
50
75
100
125
150
175
0
1000
2000
3000
4000
5000
6000
(b) Empirical Behavior Across ρ-GAM Ratio (n = 4,000, k = 40, d = 5)
LinUCB & Phased Elimination stay sublinear
despite large global ‖ε‖∞ under ρ-GAM
Final Cumulative Regret R_n
Gap-Adjusted Misspecification Parameter ρ
0
50
100
150
200
0.1
0.2
0.3
0.4
0.5
0.6